

S1 + ESAT-predecessor — Jargons & Formula Recall

S1 + ESAT-predecessor — reader-safe recall pack

Scope. This is a standalone recall pack checked against the frozen final guide. It covers the reviewed warm-up and S1 Q1/Q2/Q3/Q4/Q6/Q8. It does **not** solve S1 Q5 or Q7: they remain **[Deferred — not reviewed]**. Every glossary and formula row carries one local source/status label: [Official reference], [Lesson status], OR [Editorial recall].

Precision rule [Official reference]. On WST01/01, give **inexact decimal answers to 3 significant figures unless stated otherwise**. Exact forms are not made “more correct” by converting them to a decimal.

1. Jargon glossary

Term	Student-friendly meaning	Exact physics / maths meaning	Symbols / units	What it is NOT	Lesson example	Common trap / status
Scale-factor method [Official reference]	Change each variable by its multiplier in the original expression, then simplify. It is safer than expanding a changed bracket term-by-term.	For a product/quotient, multiply the independent scale factors; if a whole bracket scales by k , a square scales by k^2 .	Dimensionless multipliers.	Not a percentage add-on to the final answer.	NSAA: $x+y \rightarrow 1.5(x+y)$, so $(x+y)^2$ scales by $(3/2)^2$.	Use the unchanged structure first; Q17 is historical NSAA practice, not a current ESAT question.
Correlation coefficient [Official reference]	A single number describing the direction and strength of a linear association. Near +1 means strong positive linear association.	$r = S_{hw} / \sqrt{(S_{hh} S_{ww})}$, with $-1 \leq r \leq 1$.	r has no units.	Not the gradient, and not a prediction of weight.	$935/\sqrt{(286 \times 3400)} = 0.948176... \rightarrow 0.948$.	Copy the printed $S_{hh}=286$ exactly; round the inexact result to 3 s.f.
Regression line [Official reference]	A best-fit line used to predict one variable from another. Its slope must be interpreted using both variables and their units.	$w=a+bh$, $b=S_{hw}/S_{hh}$ and $a=\bar{w}-b\bar{h}$ for regression of w on h .	Here h in cm, w in kg.	Not proof of causation or an exact model for every horse.	$w=46.0+3.27h$.	State “per 1 cm, weight increases by 3.27 kg on average”; not merely “positive correlation”.
Interpolation / extrapolation [Editorial recall]	Interpolation predicts within the observed data range; extrapolation goes beyond it and is less reliable.	Classification depends on whether the input lies inside the observed x-range.	Context-dependent.	Not a property of the formula itself.	$h=153$ is interpolation; $h=170$ is extrapolation.	Use the named term and a reliability comparison.
Quartiles and IQR [Editorial recall]	Q1 and Q3 split the ordered data; the middle half lies between them. IQR measures the spread of that middle half.	$IQR=Q3-Q1$.	Same units as the data (kg here).	Not the full range and not a standard deviation.	$Q1=95$, median=105, $Q3=112$, so $IQR=17$.	The supplied outlier rule, not a memorised mean $\pm 2s$ rule, controls Q2.
Outlier fence [Official reference]	A boundary beyond which a data value is classed as unusual under the rule supplied in the question.	Lower= $Q1-1.5IQR$; upper= $Q3+1.5IQR$ when this rule is given.	Same units as data.	Not automatically an error or a value to delete.	Fences 69.5 and 137.5; 139 is the only outlier.	Say only when the question asks for the sole outlier.
Positive skew in context [Editorial recall]	A distribution has more spread on the high side of the median than on the low side.	In this box-plot comparison, $Q3-Q2 > Q2-Q1$.	Unit-free shape description.	Not merely “the right side looks longer” without context.	Club B’s weights are more varied above the median.	Name the skew and interpret it in the data context.
Discrete uniform distribution [Lesson status]	A finite list of possible values, each equally likely.	$P(Y=y)=1/6$ for $y=1,...,6$ in the die model.	Probabilities dimensionless.	Not a continuous uniform distribution.	$E(Y)=3.5$ and $E(4Y+3)=17$.	Q3 was reviewed as fully correct; do not recast it as an error.
CDF [Editorial recall]	The cumulative distribution function tells you the probability of being at or below a value.	$F(x)=P(X \leq x)$; for discrete X it is a stepwise cumulative total.	Probability 0 to 1.	Not the PMF $P(X=x)$.	At $x=0$, $F(0)=0.15$, hence $P(2X < 1)=0.15$.	Convert $2X < 1$ to $X < 0.5$, then use $X \leq 0$ because X is discrete.
Second moment and variance [Official reference]	$E(X^2)$ is the probability-weighted average of squared values; variance measures spread around the mean.	$Var(X)=E(X^2)-[E(X)]^2=\sum x^2 P(X=x)-\mu^2$.	Variance has squared units.	Not “average squares divided by the number of outcomes”; do not subtract μ instead of μ^2 .	$E(X^2)=4.6$, $Var(X)=1.36$.	This identity is printed in the formula booklet; Q4 needs no division by n .
Linear variance rule [Official reference]	Adding a constant shifts values but does not change spread; scaling by a changes variance by a^2 .	$Var(aX+b)=a^2 Var(X)$.	Variance units scale by a^2 .	Not $Var(aX+b)=a Var(X)$, and b does not enter.	$Var(5-3X)=9(1.36)=12.24 \rightarrow 12.2$.	Square the coefficient even when it is negative.
Independent events [Editorial recall]	Knowing one event occurred does not change the probability of the other.	A and B independent iff $P(A \cap B)=P(A)P(B)$.	Dimensionless.	Not mutually exclusive; independent events can overlap.	$0.24=0.60(0.33+x)$, giving $x=0.07$.	In Q6, A and C do not overlap geometrically; that does not itself establish independence.
Conditional probability [Lesson status]	Restrict attention to the cases where the condition is true, then find the fraction meeting the event.	$P(B$	$C)=P(B \cap C)/P(C)$, provided $P(C)>0$.	Dimensionless.	Not a product of two unrelated branch labels.	$0.31/(0.67+z)=31/75$ gives $z=0.08$.

Term	Student-friendly meaning	Exact physics / maths meaning	Symbols / units	What it is NOT	Lesson example	Common trap / status
Chain Venn diagram [Editorial recall]	Three circles arranged so only neighbouring sets overlap: A with B, B with C, but A never with C.	Regions: A-only, AnB, B-only, BnC, C-only, outside; no AnC or triple region.	Region areas are probabilities.	Not a three-way Venn diagram with a centre intersection.	Q6 has A-only 0.36, AnB 0.24, BnC 0.09.	Include the outside region in C' when it belongs there.
Bayes-style reversal [Editorial recall]	Reverse a tree by conditioning on the final outcome, then calculate the new branches from joint probabilities.	P(car	L)=P(car)P(L	car)/P(L).	Dimensionless.	Not an independence product.

2. Formula and case recall

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
NSAA proportional change [Official reference]	$M/M=(3/2)^2-(4/5)-(1/2)=9/10$	x,y each $\times 3/2$; z $\times 4/5$; P $\times 2$; Q unchanged	Dimensionless	A formula contains products, quotients and powers whose inputs change proportionally.	Factor a common multiplier from a bracket before applying its power.	Doubling a denominator means $\times 1/2$, not $\times 2$.	M becomes 90% of its old value: 10% decrease , option C.
Correlation [Official reference]	$r=S_{hw}/\sqrt{S_{hh}S_{ww}}$	S_{hw} cross-deviation sum; S_{hh},S_{ww} deviation sums	Unit-free	Pearson linear correlation from supplied summary statistics.	Compute each supplied sum before dividing.	0.948176... to 0.948 (3 s.f.), not 0.95.	$935/\sqrt{286 \times 3400}=0.948$.
Regression of w on h [Official reference]	$w=a+bh$; $b=S_{hw}/S_{hh}$; $a=\bar{w}-b\bar{h}$	a intercept; b gradient; h height; w weight	b: kg cm^{-1} ; a: kg	Predict weight from height with the stated regression direction.	$a=520-3.27(145)=46.0$.	Do not swap variables without changing the regression.	$w=46.0+3.27h$; $h=153$ gives 546 kg.
Box-plot outlier rule (QP) [Official reference]	$IQR=Q3-Q1$; fences $Q1-1.5IQR$, $Q3+1.5IQR$	Q1 lower quartile; Q3 upper quartile	kg here	Only when this rule is supplied by the question.	$1.5 \times 17=25.5$.	Do not substitute a $\text{mean} \pm 2s$ rule.	$95-25.5=69.5$; $112+25.5=137.5$; 139 only.
Discrete expectation [Official reference]	$E(g(X))=\sum g(x)P(X=x)$	Sum over all possible values	Units of g(X)	Expectation of a transformed discrete variable.	$E(4Y+3)=4E(Y)+3$.	Keep probabilities with their outcomes.	$E(Y)=3.5 \Rightarrow E(4Y+3)=17$.
CDF / threshold [Editorial recall]	$F(x)=P(X \leq x)$	F is cumulative probability	Unit-free	A question asks for a “ $\leq$ ” probability or a transformed discrete threshold.	$2X < 1 \Leftrightarrow X < 0.5 \Leftrightarrow X \leq 0$.	F is cumulative, not a single PMF bar.	$F(0)=0.05+0.10=0.15$.
Variance from a PMF [Official reference]	$\text{Var}(X)=\sum x^2P(X=x)-\mu^2$	$\mu=E(X)$	Squared X-units	Discrete distribution with probabilities and a mean.	$E(X^2)=4.6$; $\mu^2=1.8^2=3.24$.	No division by number of listed values.	$\text{Var}(X)=4.6-3.24=1.36$.
Linear variance [Official reference]	$\text{Var}(aX+b)=a^2\text{Var}(X)$	a multiplier; b shift	Squared output units	Spread after a linear transformation.	b disappears; square a.	Negative a still gives positive a^2 .	$\text{Var}(5-3X)=9 \times 1.36=12.24=\mathbf{12.2}$ (3 s.f.).
Independence in a Venn diagram [Official reference]	$P(A \cap B)=P(A)P(B)$	$P(A)=0.60$; $P(B)=0.33+x$	Unit-free	Independence is explicitly stated.	$0.24/0.60=0.33+x$.	Do not count BnC in BnC'.	$x=0.07$, so $P(B)=0.40$.
Conditional probability [Lesson status]	P(B	C')= $P(B \cap C')/P(C')$	C' complement of C	Unit-free	“Given” is stated; denominator is the condition.	$0.31/(0.67+z)=31/75$.	All of C', including outside z, belongs in the denominator.
Total probability / Bayes [Editorial recall]	$P(L)=\sum P(\text{mode})P(L$	mode); P(car	L)=P(car)P(L	car)/P(L)	L=late	Unit-free	Tree paths or reversal of a conditional tree.
Reversed tree check [Editorial recall]	$\sum \text{branches given } L=1$ and $\sum \text{branches given } L'=1$	Branches condition on the parent node	Unit-free	A tree is reversed.	Given L': $6/13+9/52+19/52=1$.	Do not mix the late and not-late fans.	Given L: $3/4+1/8+1/8=1$.

3. Coverage and boundary checklist

Target	Reader-safe coverage	Boundary
NSAA Section 1 specimen, Part A Q17 S1 Q1 S1 Q2 S1 Q3 S1 Q4 S1 Q5 / Q7 S1 Q6 S1 Q8	Scale factors, multiplier 9/10, 10% decrease / option C $r=0.948$; regression, contextual gradient, interpolation/extrapolation IQR 17, fences 69.5 / 137.5, only outlier 139, skew interpretation Discrete uniform and $E(4Y+3)=17$ PMF/CDF, $\text{Var}(X)=1.36$, $\text{Var}(5-3X)=12.2$ No solution content included Chain Venn, $x=0.07$, $z=0.08$, $y=0.16$, $P(B)=0.40$ Forward and reversed tree, $P(L)=2/15$, $P(\text{car}$	Historical ESAT-predecessor practice; not a current ESAT question . Reviewed; exact paper facts are [Official reference]. Reviewed; supplied $1.5 \times \text{IQR}$ rule only. Reviewed as fully correct. Q4—not legacy Q3 numbering. [Deferred – not reviewed]; attempt unaided before a later revisit. Q6(b) is a tutor teaching moment, not a learner-error claim. $L)=3/4$

4. Content provenance

- Content check:** frozen final guide SHA-256 23c096cad3bb3948b58a0b02888db380c0961be00aadd0129a711ae7aa9bc327.
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